HW10 for CS 6210

You may (and probably should) talk about problems with the each other, with the TA, and with me, providing attribution for any good ideas you might get. Your final write-up should be your own.

```
1 using LinearAlgebra
```

1. Your turn

Submit a course evaluation! I can see if you have submitted an evaluation (even if I don't see what it is). This is worth 2 points on the final grade.

2. Scaled iteration

Suppose $A = QR$ is an economy QR decomposition for $A \in \mathbb{R}^{m \times n}$ a tall thin matrix of full column rank. Consider the stationary iteration

$$x^{(k+1)} = R^{-1}(Q^T b - \lambda R^{-T} x^{(k)})$$

Suppose $x^{(0)} = 0$. Then write the other iterates $x^{(k)}$ in the form $x^{(k)} = V f_k(\Sigma) U^T b$ where $A = U\Sigma V^T$ is the economy SVD. For what range of λ does the iteration converge?

Note that $A^T A = R^T R$, so $R^T R$ is symmetric positive definite (SPD) since A has full column rank. The iteration can be rewritten as $x^{(k+1)} = s - \lambda P x^{(k)}$ where $s = R^{-1} Q^T b$ and $P = R^{-1} R^{-T}$.

Can verify that $P = (A^T A)^{-1} = (R^T R)^{-1}$ because $P^{-1} = (R^{-1} R^{-T})^{-1} = R^T R = A^T A$ Also, $s = (A^T A)^{-1} A^T b$, which is the least-squares solution to $Ax = b$.

Starting from $x^{(0)} = 0$:

$$x^{(1)} = s$$

$$x^{(2)} = s - \lambda P s$$

$$x^{(3)} = s - \lambda P(s - \lambda P s) = s - \lambda P s + \lambda^2 P^2 s$$

and in general $x^{(k)} = \sum_{j=0}^{k-1} (-\lambda)^j P^j s$

use the SVD $A = U\Sigma V^T$. Then $A^T A = V\Sigma^2 V^T, \quad P = (A^T A)^{-1} = V\Sigma^{-2} V^T$, $P^j = V\Sigma^{-2j} V^T$

Also $A^T b = V\Sigma U^T b, \quad s = V\Sigma^{-1} U^T b$

Thus $P^j s = V\Sigma^{-2j}V^T \cdot V\Sigma^{-1}U^T b = V\Sigma^{-2j-1}U^T b$, $x^{(k)} = V\left(\sum_{j=0}^{k-1}(-\lambda)^j\Sigma^{-2j-1}\right)U^T b$

The term in parentheses is $\sum_{j=0}^{k-1}(-\lambda)^j\Sigma^{-2j-1} = \Sigma^{-1}\sum_{j=0}^{k-1}\left(-\lambda\Sigma^{-2}\right)^j$ (Operations are elementwise on the diagonals.) Thus, $f_k(\Sigma) = \Sigma^{-1}\sum_{j=0}^{k-1}\left(-\lambda\Sigma^{-2}\right)^j$

For convergence as $k \to \infty$, the infinite series $\sum_{j=0}^{\infty} r_i^j$ must converge for each ratio $r_i = -\lambda/\sigma_i^2$, requiring $|r_i| < 1$ for all i, i.e., $\left|\frac{\lambda}{\sigma_i^2}\right| < 1 \quad \forall i$

This holds if $|\lambda| < \min_i \sigma_i^2 = \sigma_n^2$ (the smallest singular value squared).

3. Rational roots

Consider a rational function of the form

$$f(z) = c^T(zI - A)^{-1}b + d$$

Write a routine `rateval(z, F, b, c, d)` to compute $f(z)$ in $O(n^2)$ time given the precomputed Hessenberg factorization from `F = hessenberg(A)`.

Note that we can write this by as $f(z) = c^T u$ where $u = (zI - A)^{-1}b$, or equivalently

$$(A - zI)u + b = 0.$$

Putting it all together, we have

$$\begin{bmatrix} A - zI & b \\ c^T & d \end{bmatrix}\begin{bmatrix} u \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ f(z) \end{bmatrix}.$$

The function call `eigvals(A,B)` finds all solutions of the eigenvalue problem $Ax = \lambda Bx$. Using this function, write a routine `ratroots(A, b, c, d)` to compute all the *real* zeros of f.

You should check correctness of the roots returned by `ratroots` using the fast evaluation code from `rateval`.

```
rateval (generic function with 1 method)
```

```
function rateval(z, F, b, c, d)
    H = F.H
    Q = F.Q
    n = size(H, 1)
    qb = Q' * b
    y = (z * I(n) - H) \ qb
    x = Q * y

    dot(c, x) + d
end
```

```
ratroots (generic function with 1 method)
```

```julia
function ratroots(A, b, c, d)
    n = size(A, 1)
    M = [A b; c' d]
    B = [I(n) zeros(n,1); zeros(1,n) 0.0]
    zs = eigvals(M, B)
    real_zs = [real(z) for z in zs if isfinite(z) && isreal(z) && abs(imag(z)) < 1e-10]

    real_zs
end
```

rateval: The function $f(z) = c^T(zI - A)^{-1}b + d$ requires solving $(zI - A)u = b$ to compute $f(z) = c^Tu + d$, using the precomputed Hessenberg factorization F where $F.Q'AF.Q = F.H$ (upper Hessenberg). To achieve O(n²) time:

Compute $rhs = F.Q'b$. Solve $(zI - H)v = rhs$ using Hessenberg factorization. Compute $u = F.Qv$, then $f(z) = c^Tu + d$.

ratroots: The real zeros of $f(z) = 0$ are the real finite generalized eigenvalues of the pencil (M, B), where: $M = \begin{bmatrix} A & b \\ c^T & d \end{bmatrix}$, $B = \begin{bmatrix} I_n & 0 \\ 0^T & 0 \end{bmatrix}$. This follows from $\det(M - zB) = 0$ implying $f(z) = 0$ (Schur complement or block determinant). Compute using eigvals(M, B), filter real and finite values, and verify with rateval